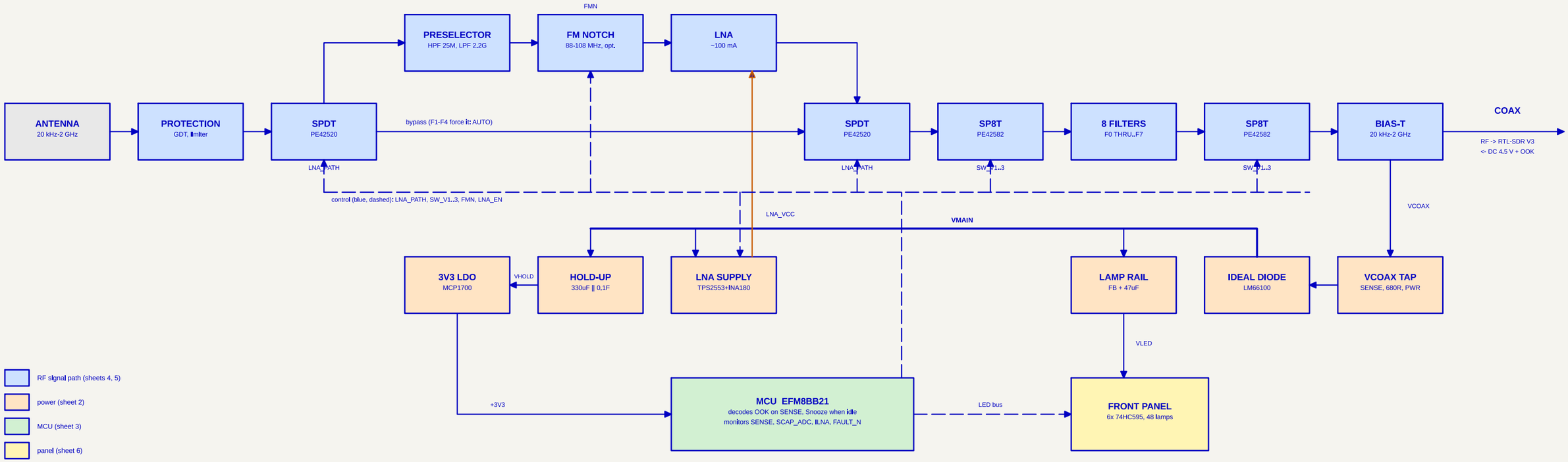
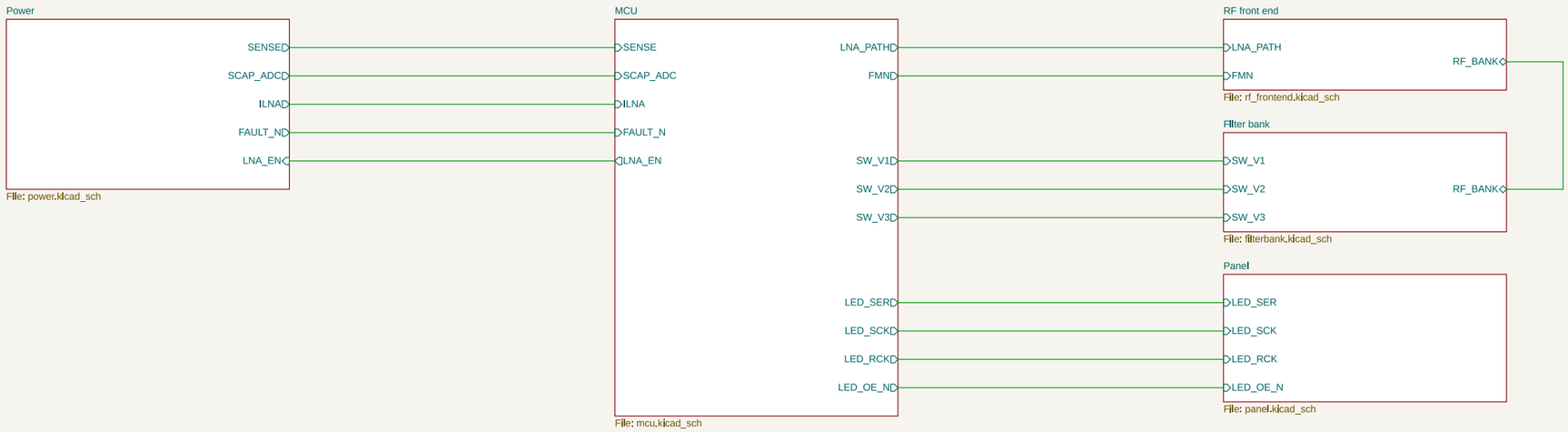
Coaxolotl - block diagram



Sheets



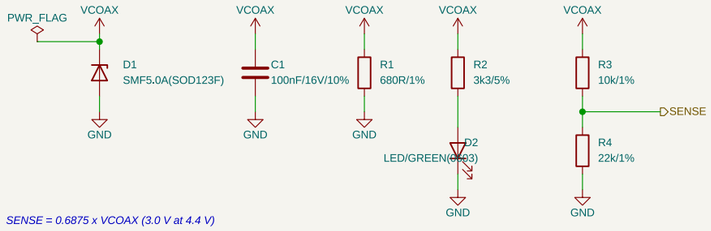
Power and ground cross sheets only as power symbols (VCOAX, VMAIN, VHOLD, VLED, LNA_VCC, +3V3, GND).

Signals cross sheets as hierarchical labels through the sheet pins above. Firmware pin map: firmware/src/hal_efm8bb2.c

Generated by hardware/kicad/tools/build.py — edit the Python, not this file
Bias-T powered, OOK switched LNA + 8-way filter bank
z2labs

Sheet: /		
File: coaxolotl.kicad_sch		
Title: Coaxolotl		
Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/6

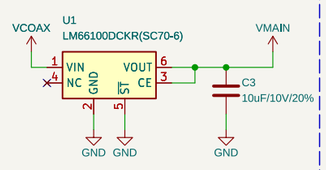
VCOAX tap



SENSE = 0.6875 x VCOAX (3.0 V at 4.4 V)

t_fall = 1.1 x R_eff x C_total: R1 sets the OOK falling edge

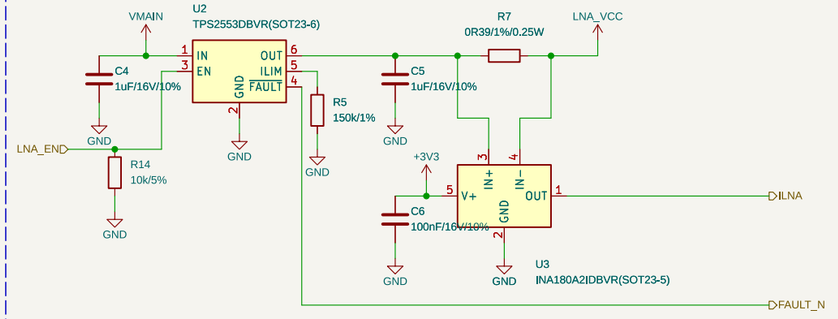
Ideal diode



CE = VOUT: comparator blocks

reverse current in OOK gaps

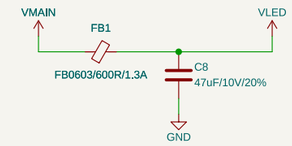
LNA supply + current monitor



Ilim = f(R5), TPS2553 Eq.1: VERIFY IOS(min) > 130 mA

ILNA = 0.39R x 50 V/V = 19.5 mV/mA (100 mA -> 1.95 V); Kelvin taps at R7 pads

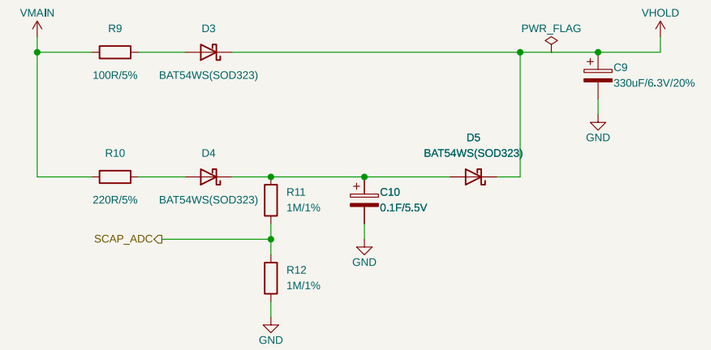
Lamp rail



48 lamps x 1 mA max on VMAIN:

dark in OOK gaps, no hold-up drain

Hold-up (diode-OR)

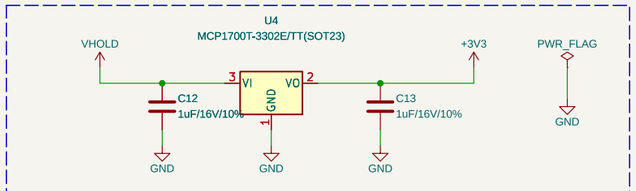


fast path R9/D3: VHOLD up in ~30 ms (100R x 330uF) -> MCU boots

slow path R10/D4: tau ~22 s (220R x 0.1F); ride-through ~4 min at 0.6 mA

SCAP_ADC = VSCAP / 2 (HOLD lamp at >= 3.6 V)

3V3 regulator



MCP1700: Iq 1.6 uA, dropout 178 mV;

VO needs >= 1 uF ceramic

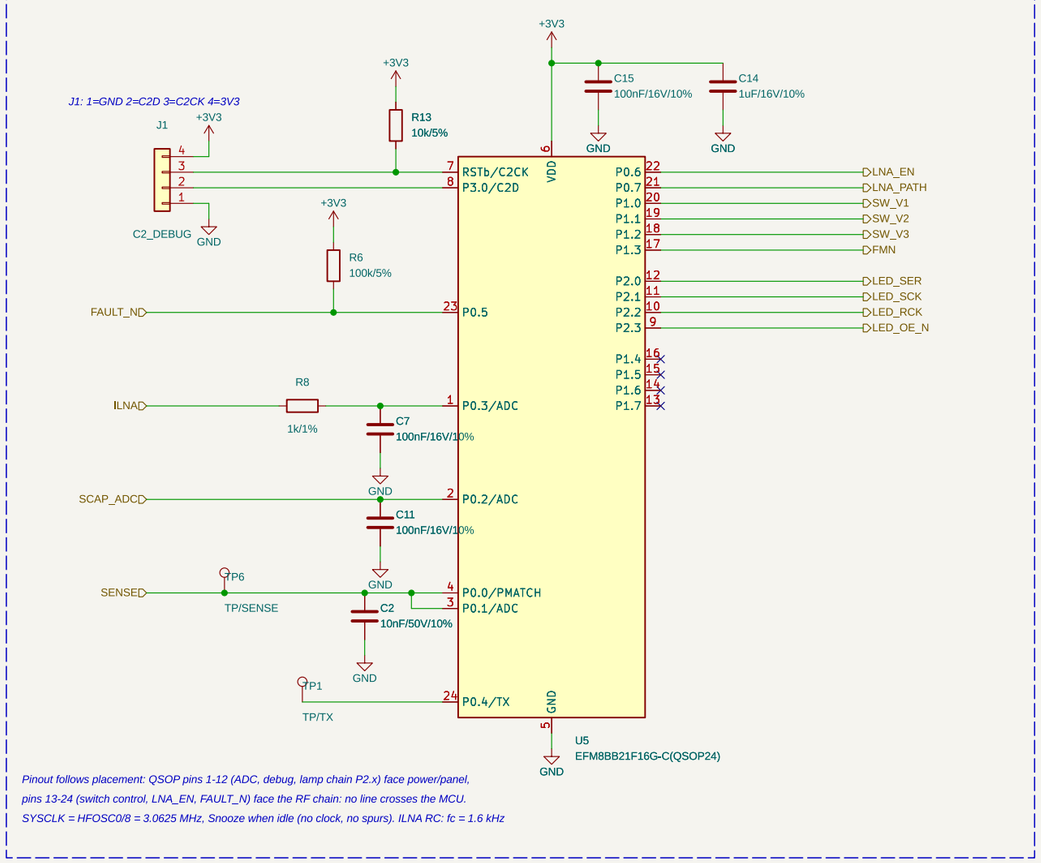
Generated by hardware/kicad/tools/build.py — edit the Python, not this file

VCOAX tap, ideal diode, LNA supply switch + current monitor, lamp rail, hold-up, 3V3

z2labs

Sheet: /Power/		
File: power.kicad_sch		
Title: Power		
Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/6

MCU



Pinout follows placement: QSOP pins 1-12 (ADC, debug, lamp chain P2.x) face power/panel, pins 13-24 (switch control, LNA_EN, FAULT_N) face the RF chain: no line crosses the MCU.

SYSCLK = HFOSC0/8 = 3.0625 MHz, Snooze when idle (no clock, no spurs). ILNA RC: fc = 1.6 kHz

Generated by hardware/kicad/tools/build.py — edit the Python, not this file

EFM8BB21F16G, C2 debug, ADC input conditioning

z2labs

Sheet: /MCU/

File: mcu.kicad_sch

Title: MCU

Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/6

Antenna input + protection

J2
SMA/EDGE(ANT)

GND
GDT1
GDT/90V

D6
BAV99W(SOT323)

R23
100k/5%

GDT 90 V, BAV99 low-C clamp, 100k bleed

LNA/BYP IN

+3V3
C36
100nF/16V/10%
GND

RFC, VDD, LS, RF1, CTRL, GND, RF2

U6
PE42520C-Z(QFN16)

bypass: F1-F4 (HF/LF) always, LNA off or tripped

LNA/BYP OUT

+3V3
C37
100nF/16V/10%
GND

RF1, LS, VDD, RFC, RF2, GND, CTRL

U7
PE42520C-Z(QFN16)

RF_BANK

LNA

LNA_VCC
C40
100pF/C0G
GND
GND
C32
100pF/C0G
FB2
FB0603/600R/1.3A
R25
3k32/1%
L4
560nH/5%
GND
C30
100pF/C0G
L8
18nH/2%
C28
1nF/50V/C0G
RF_IN, VBIAS, RF_OUT, SHDN, GND
U13
QPL9547TR7(DFN8-2x2)
C31
1nF/50V/C0G
GND
GND

SWL434 stage 1
+ 18 nH (EVB L1)
1 nF DC blocks

Preselector (LNA path only)

HPF: 5th-order Butterworth T, fc ~25 MHz
LPF: LTCC, fc ~2.2 GHz

C24
200pF/50V/C0G
C25
68pF/50V/C0G
L1
220nH/5%
GND
C26
200pF/50V/C0G
L2
220nH/5%
GND
U12
LFCN-2250+(FV1206)
IN, GND, OUT
GND
R22
0R/0603

FM notch (DNP option)

+3V3
RFC, VDD, LS, RF1, CTRL, GND, RF2
GND
U9
NA(PE42520C-Z)

L3
NA(56nH/2%)
C27
NA(47pF/C0G)

+3V3
RF1, LS, VDD, RFC, RF2, GND, CTRL
GND
U10
NA(PE42520C-Z)

tank 56 nH || 47 pF: f0 = 98 MHz

Control inputs (reset-safe pull-downs)

FMN
R19
10k/5%
GND

LNA_PATH: 0 = RF1 = bypass, 1 = RF2 = LNA. FMN: 0 = RF1 = through, 1 = RF2 = notch.
PE42520 RF ports sit at 0 VDC: no blocking caps except around the LNA.

LNA_PATH
R15
10k/5%
GND

Generated by hardware/kicad/tools/build.py — edit the Python, not this file
Antenna protection, LNA/bypass select, preselector, FM notch option, LNA
z2labs

Sheet: /RF front end/
File: rf_frontend.kicad_sch

Title: RF front end

Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/6

Filter bank

RF_BANK
SW_V1
SW_V2
SW_V3
+3V3
GND
C38 100nF/16V/10%
R21 0R/0603
U8 PE42582A(QFN24)
U11 PE42582A(QFN24)
C39 100nF/16V/10%

F1 LPF 550 kHz
L101 18uH/5%
L102 18uH/5%
C101 5.6nF/50V/C0G
C102 10nF/50V/C0G
C103 5.6nF/50V/C0G

F2 BPF 1.6-4 MHz
C201 2.2nF/50V/C0G
C202 1.2nF/50V/C0G
C203 2.2nF/50V/C0G
L203 2.2uH/5%
L204 2.2uH/5%
L201 4.7uH/5%
L202 4.7uH/5%
C204 820pF/50V/C0G
C205 1.5nF/50V/C0G
C206 820pF/50V/C0G

F3 BPF 4-12 MHz
C301 820pF/50V/C0G
C302 470pF/50V/C0G
C303 820pF/50V/C0G
L303 820nH/5%
L304 820nH/5%
L301 1.5uH/5%
L302 1.5uH/5%
C304 270pF/50V/C0G
C305 470pF/50V/C0G
C306 270pF/50V/C0G

F4 BPF 12-30 MHz
C401 270pF/50V/C0G
C402 150pF/50V/C0G
C403 270pF/50V/C0G
L403 330nH/5%
L404 330nH/5%
L401 560nH/5%
L402 560nH/5%
C404 100pF/50V/C0G
C405 180pF/50V/C0G
C406 100pF/50V/C0G

F5 BPF 30-80 MHz
C501 100pF/50V/C0G
C502 56pF/50V/C0G
C503 100pF/50V/C0G
L503 150nH/5%
L504 150nH/5%
L501 220nH/5%
L502 220nH/5%
C504 47pF/50V/C0G
C505 82pF/50V/C0G
C506 47pF/50V/C0G

F6 BPF 108-174 MHz
C601 27pF/50V/C0G
C602 15pF/50V/C0G
C603 27pF/50V/C0G
L603 56nH/5%
L604 56nH/5%
L601 56nH/5%
L602 56nH/5%
C604 18pF/50V/C0G
C605 33pF/50V/C0G
C606 18pF/50V/C0G

F7 HPF 380 MHz
C701 8.2pF/50V/C0G
C702 4.7pF/50V/C0G
C703 8.2pF/50V/C0G
L701 15nH/5%
L702 15nH/5%

Bias-T + coax to RTL-SDR

VCOAX
L7 1mH/0.2A
L6 10uH/0805
R24 220R/5%
L5 100nH/0603
J3 SMA/EDGE(RTL-SDR)
C33 2.2uF/16V/X7R
C34 1nF/50V/C0G
C35 22pF/50V/C0G

DC block 2.2 uF ‖ 1 nF ‖ 22 pF; DC feed 100 nH -> 10 uH ‖ 220R -> 1 mH:
20 kHz-2 GHz. VERIFY S21/S11 with a VNA. D1 (TVS) sits at J3 (power sheet).

Control inputs

SW_V1
SW_V2
SW_V3
R16 10k/5%
R17 10k/5%
R18 10k/5%

Branch k on RFk+1 of both switches. LS = 0 code (V3 V2 V1):
RF1 000, RF2 100, RF3 010, RF4 110, RF5 001, RF6 101, RF7 011, RF8 111
(mapped in tools/gen_bandplan.py). V4 = LS = VSS_EXT = GND.
Filters: 5th-order Chebyshev 0.1 dB, 50 ohm, E12 values,
BPF = HPF + LPF (tools/filters.py). Ideal-part response there;
VERIFY with real L Q / SRF and the layout on a VNA.

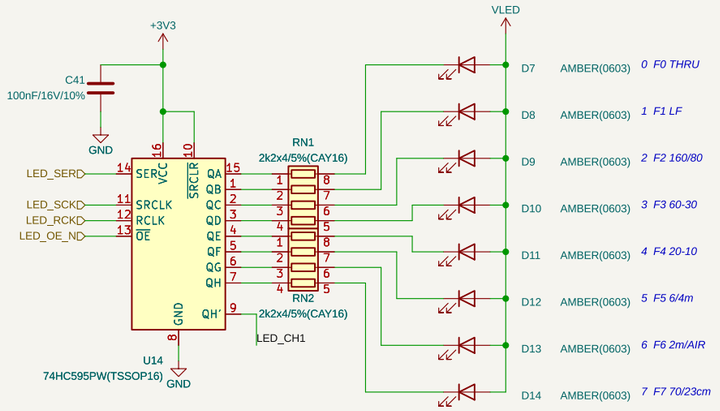
Generated by hardware/kicad/tools/build.py — edit the Python, not this file
2x PE42582 SP8T, 8 filter branches, bias-T, coax
z2labs

Sheet: /Filter bank/
File: filterbank.kicad_sch

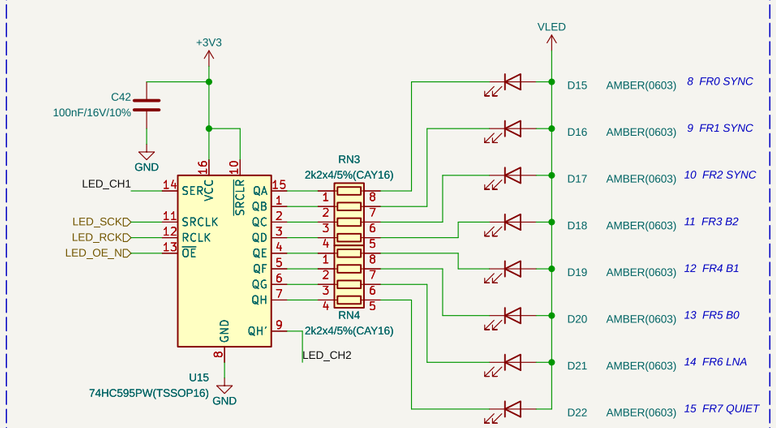
Title: Filter bank

Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/6

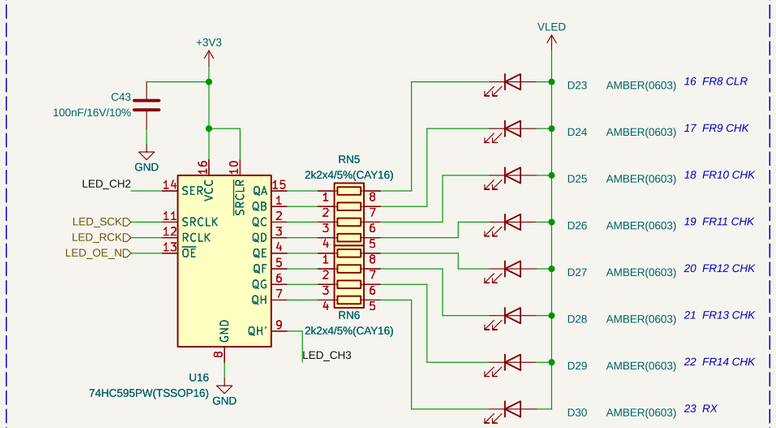
BANK (lamps 0-7)



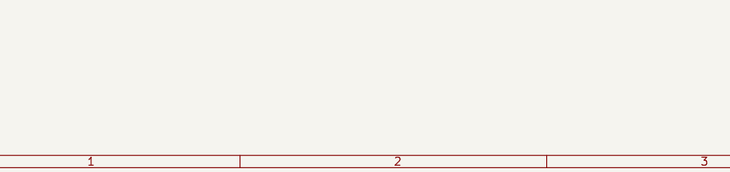
FRAME (lamps 8-15)



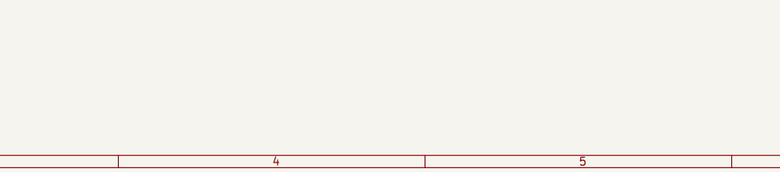
FRAME + RX (lamps 16-23)



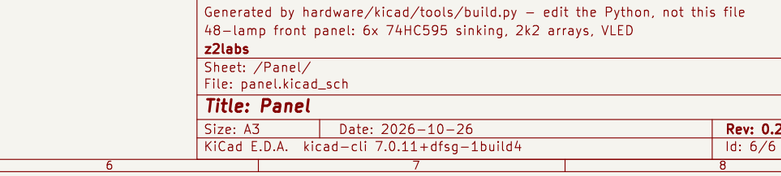
DECODE (lamps 24-31)



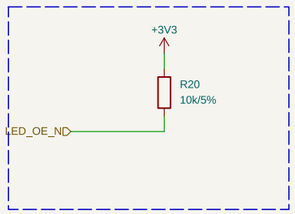
DECODE + RF (lamps 32-39)



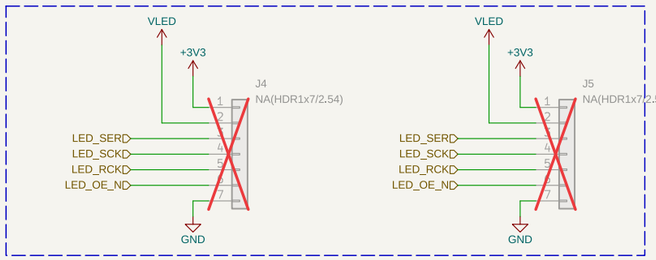
RF + SYSTEM (lamps 40-47)



OE_N pull-up



Snap-off lamp strip link



Lamp n: register n/8 (U14 = first after the MCU), output Q(n%8) - firmware/src/panel.h.
Q low = lamp on (sink). 1 mA per lamp: (VLED - 2 V - 0.1 V) / 2k2. Use 2 V LEDs only.
Lamp test lights all 48 (48 mA) with the LNA still off. Outside it, at most 23 lamps.
Panel modes (config frame, kept in flash): full / bands = U14 only / off = OE_N high.
U14-U19, RN1-RN12, D7-D54, C41-C46 and J5 sit on the snap-off strip; J4 on the main board.

Generated by hardware/kicad/tools/build.py — edit the Python, not this file
48-lamp front panel: 6x 74HC595 sinking, 2k2 arrays, VLED

z2labs

Sheet: /Panel/		
File: panel.kicad_sch		
Title: Panel		
Size: A3	Date: 2026-10-26	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/6